



United International University

School of Science and Engineering

Final Examination Trimester: Spring-2024

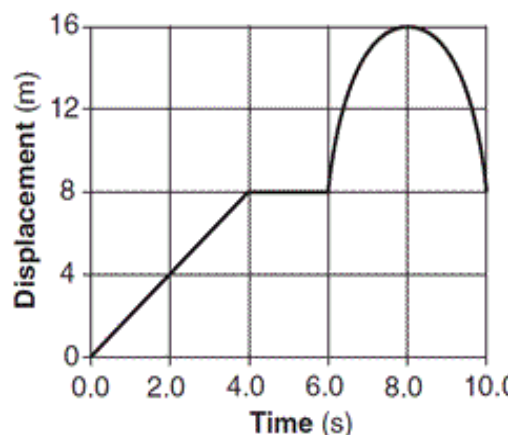
Course Title: Fundamental Calculus

Course Code: Math 1151 Marks: 40 Time: 2 Hours

Answer all the questions. Answer all parts of a question together.

1. (a) The following figure represents a position function of a particle at time t . [5]

- (i) Find the average velocity over the time interval $[2, 6]$.
- (ii) What is the instantaneous velocity at $t = 8$. Explain the reason.
- (iii) Determine whether the velocity doesn't exist. Explain the reason.
- (iv) Roughly sketch the velocity graph of the particle.



- (b) Consider the function [5]

$$f(x) = \begin{cases} 3 - x^2; & x < 1 \\ 4 - 2x; & x \geq 1 \end{cases}$$

- (i) Sketch the graph of $f(x)$.
- (ii) Determine whether the function $f(x)$ is continuous and differentiable at $x = 1$.

2. (a) Consider, the function $f(x) = x^2 - 2$. [1]

- (i) Find the slope of tangent line to the graph of $f(x)$ at the point $x = 0$. [1]
- (ii) Find the equation tangent line to the graph of $f(x)$ at the point $x = 0$. [2]
- (iii) Draw the graph of $f(x)$ together with the tangent line from (ii) in the same axes. [2]

- (b) (i) If $s = \cot w$ and $w = t^3 - \frac{2}{\sqrt{t}} + 3$, then find $\frac{ds}{dt}$. [2.5]

- (ii) If $y = \sec^3(2 - 3x^2 \cos x)$, then find $\frac{dy}{dx}$. [2.5]

3. (a) Evaluate the following integrals:

(i) $\int \frac{t^3 - t^2 \cos t - 2t + 3}{t^2} dt$ [2]

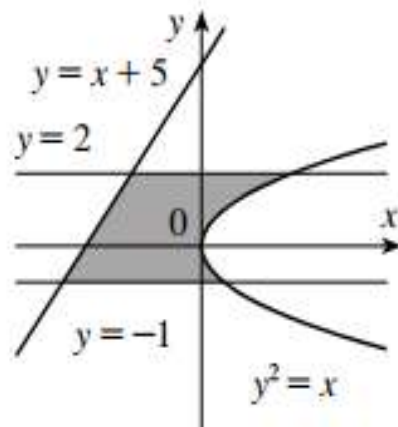
(ii) $\int \frac{x^3}{\sqrt{5-2x^4}} dx$ [2]

(iii) $\int \frac{x}{1+x^4} dx$ [2]

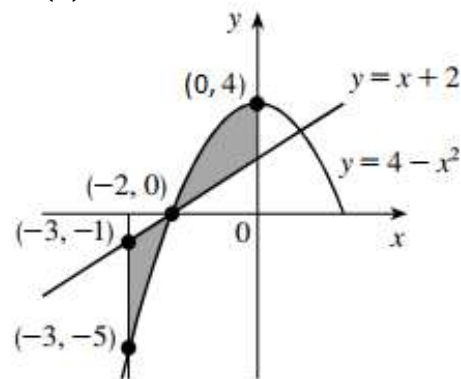
(b) Evaluate $\int_{-2}^4 f(x) dx$, where $f(x) = \begin{cases} 2x + 1; & x \leq 1 \\ 5 - 2x; & x > 1 \end{cases}$ [4]

4. (a) Use a suitable method to **evaluate** the area of the following shaded regions: [7]

(i)



(ii)



(b) Evaluate the integral: $\int_0^{\pi} 2t \sin 2t dt$ [3]